



以下有一份完整的 JD :

Senior AI Algorithms Software Engineer, Time Series/Forecasting

NVIDIA is searching for a Senior AI & Algorithms Engineer to develop time-series and temporal modeling algorithms for AI applications in manufacturing and industrial systems. AI is transforming how continuous sensor data is collected, monitored, and analyzed for use cases such as forecasting, anomaly detection, and predictive maintenance. NVIDIA Metropolis for Factories and Manufacturing is leading this transformation with scalable, production-ready AI platforms.

This role requires deep expertise in machine learning and deep learning for time-series data, along with hands-on experience building real-world AI systems.

What You'll Be Doing :

Design and develop time-series ML/DL algorithms for forecasting, anomaly detection, and temporal pattern analysis.

Work with large-scale, multivariate, and noisy temporal datasets from sensors and event streams.

Build and optimize end-to-end pipelines for training and deploying time-series models in production.

Collaborate with cross-functional software, research, and hardware teams.

Provide technical leadership and mentor junior engineers.

What We Need To See :

3+ years of industry experience with an MS or PhD in a relevant field.

Strong background in time-series analysis, statistics, and signal processing.

Proficiency in PyTorch and/or TensorFlow, and strong Python or C++ skills.

Proven ability to deliver production-quality AI solutions.

Ways To Stand Out From The Crowd :

Experience with real-time or streaming time-series systems.

Knowledge of probabilistic forecasting or uncertainty estimation.

Familiarity with MLOps and deployment at scale.

Experience designing or building agentic AI systems.

Contributions to open-source or applied research.

NVIDIA is widely considered to be one of the technology world's most desirable employers. We have some of the most forward-thinking and hardworking people in the world working for us. Are you a creative and autonomous data scientist who loves challenges? Do you have a genuine passion for advancing the state of AI & machine learning across a variety of industries? If so, we want to hear from you. Come and join our Metropolis / team where you'll help build our real-time, cost-effective computing platform driving our success in this exciting and quickly growing field.

NVIDIA is committed to fostering a diverse work environment and proud to be an equal opportunity employer. As we highly value diversity in our current and future employees, we do not discriminate (including in our hiring and promotion practices) on the basis of race, religion, color, national origin, gender, gender expression, sexual orientation, age, marital status, veteran status, disability status or any other characteristic protected by law.

請根據以上內容，完整列出這份工作 "所有技術技能"，我要的是技能細項，不是大方向技能樹而已。

例如："transformer、LSTM..."、"Stationary、Heterogeneous..."等等

我希望你完整列出來，我想要針對你這份整理的清單主題，做一個完整的複習

熊熊，先說結論：這個 JD 隱含的「技術技能細項」非常多，可以拆成幾大塊：

時間序列理論 / 統計 / 訊號處理、ML/DL 模型與訓練技巧、Forecasting / Anomaly / Predictive Maintenance domain 知識、資料與系統工程、MLOps / 部署 / 即時串流、Probabilistic / Uncertainty、Agentic AI 等。

下面直接用條列，把你之後可以一項項複習的主題列出來。

1. 時間序列分析／統計／訊號處理相關技能

- 時間序列基本性質
 - Stationarity (平穩性：嚴格平穩 / 弱平穩)
 - Non-stationary (非平穩)
 - Trend (趨勢)
 - Seasonality (季節性)
 - Cyclic behavior (週期性)
 - Heteroskedasticity / Heterogeneous variance (異質變異 / 時變變異)
 - Autocorrelation / Partial autocorrelation (自相關 / 偏自相關)
 - Cross-correlation (多變量序列之間的互相關)
- 傳統時間序列模型
 - AR / MA / ARMA
 - ARIMA / SARIMA / SARIMAX
 - VAR / VARMAX
 - State-space models (狀態空間模型)
 - Kalman filter / Extended Kalman filter
- 統計檢定與估計
 - 單位根檢定 (ADF test, KPSS test)
 - Stationarity 檢定、白噪音檢定
 - Cointegration (共整合)
 - Granger causality (Granger 因果)
 - Maximum likelihood estimation (極大似然估計)

- Bayesian estimation (貝氏估計) 基礎
- 資料特性與前處理
 - Detrending (去趨勢) 、Deseasonalization (去季節性)
 - Differencing (差分)
 - Normalization / Standardization / Robust scaling
 - Handling missing data (缺失值填補 : forward-fill、interpolation、model-based)
 - Outlier detection (統計式異常點偵測)
 - Resampling / Aggregation (down-sampling / up-sampling)
 - Irregular time series handling (不規則採樣序列處理)
 - Sliding window / rolling window 轉成 supervised data
- 訊號處理 (特別是感測器資料)
 - Sampling theory (取樣理論、Nyquist 頻率)
 - Filtering (低通 / 高通 / 帶通 / 卡爾曼濾波)
 - FFT / STFT / Wavelet transform (頻域特徵)
 - SNR (Signal-to-Noise Ratio) 與去雜訊 (denoising)

2. 機器學習與深度學習 (時間序列導向)

- 一般機器學習基礎
 - Supervised / Unsupervised / Semi-supervised learning
 - Regression / Classification / Clustering
 - Cross-validation (時序型 CV : rolling / expanding window)
 - Regularization (L1 / L2 / Elastic net)
 - Feature selection / Feature engineering for time series
- 時間序列深度學習模型
 - RNN (Recurrent Neural Networks)
 - LSTM (Long Short-Term Memory)
 - GRU (Gated Recurrent Unit)
 - Temporal Convolutional Network (TCN)
 - 1D-CNN for time series
 - Transformer for time series
 - Self-attention / Multi-head attention
 - Encoder-only / Encoder-decoder 架構
 - Positional / Temporal encoding
 - Sequence-to-sequence (Seq2Seq) models

- Encoder-Decoder with attention (用於 forecasting / anomaly)
- 表徵學習與進階架構
 - Multivariate time-series embedding (多變量序列嵌入)
 - Graph-based temporal models (例如 T-GNN, DCRNN 等)
 - Temporal point process / event sequence modeling
 - Autoencoder / VAE for time-series (壓縮表徵、異常偵測)
 - Contrastive learning for time series (self-supervised TS 表徵)
- 訓練與優化技巧
 - Loss functions for time series
 - MSE / MAE / Huber loss
 - Quantile loss (for probabilistic forecasting)
 - Custom anomaly-detection loss (e.g., reconstruction error, ranking loss)
 - Regularization : dropout、weight decay、early stopping
 - Sequence length / receptive field 設計
 - Batching strategies for sequences (padding / masking / bucketing)
 - Handling covariates (static features / future known features)

3. Forecasting (時間序列預測) 專項技能

- 預測問題定義
 - Point forecasting (點預測)
 - Multi-step forecasting (多步預測, recursive / direct / multi-output)
 - Long-horizon forecasting (長視野預測)
 - Global vs local models (全域模型 vs per-series 模型)
- 特定 forecasting 技巧與概念
 - Multi-horizon forecasting (同時預測多個未來時間點)
 - Multi-variate forecasting (多變量預測)
 - Hierarchical / grouped time series forecasting
 - Probabilistic forecasting (機率式預測)
 - Predictive distribution modeling
 - Quantile regression / Pinball loss
 - Prediction intervals / confidence intervals
 - Backtesting / rolling origin evaluation
 - Forecast accuracy metrics
 - MAPE, sMAPE, RMSE, MAE, MASE, CRPS, coverage

4. 異常偵測 (Anomaly Detection) 與預知維護 (Predictive Maintenance)

- 異常偵測方法
 - Reconstruction-based (Autoencoder, VAE, Seq2Seq, Transformer AE)
 - Density-based (e.g., KDE, Isolation Forest, One-class SVM)
 - Distance / clustering-based (kNN, clustering outliers)
 - Forecasting residual-based anomaly detection
- 時間序列異常偵測細節
 - Point anomalies / contextual anomalies / collective anomalies
 - Online vs offline anomaly detection
 - Thresholding strategies (統計門檻 / learned threshold)
 - Handling concept drift (概念飄移)
- 預知維護 (Predictive maintenance)
 - Remaining Useful Life (RUL) estimation
 - Health index / degradation modeling
 - Event prediction (早期故障預警)
 - Feature construction from sensor data (頻譜特徵、統計特徵、trend features)

5. 感測器與工業時間序列資料處理

- 數據來源與型態
 - Multivariate sensor data (多元感測器)
 - Event logs / machine states / alarms
 - Time-stamped event streams (event time vs processing time)
- 工業資料典型問題
 - Noisy data (高雜訊)
 - Missing / corrupt sensor readings
 - Different sampling rates (異步感測器)
 - Sensor drift / calibration issues
- 特徵工程
 - Statistical features (mean, std, skewness, kurtosis, peak-to-peak)
 - Rolling window features (rolling mean, rolling std, rolling max/min)
 - Frequency domain features (power spectral density, band power)
 - Cross-sensor interaction features (correlations / lags)

6. 軟體工程與程式語言技能

- 語言與框架
 - Python (核心)
 - C++ (高效實作、可能與 CUDA / GPU stack 整合)
 - PyTorch
 - TensorFlow / Keras
- 典型 Python 生態系
 - NumPy, SciPy
 - Pandas (時間序列 index、resample、rolling)
 - Scikit-learn (傳統 ML、前處理、metrics)
 - Statsmodels (傳統 TS 模型：ARIMA、VAR 等)
- 效能與最佳化
 - Vectorization (向量化運算)
 - 多執行緒 / 多處理 (Python threading / multiprocessing)
 - C++ extension / PyBind11 (如要和 C++ 整合)
 - GPU 加速概念 (batching、memory layout)

7. End-to-End Pipeline 與系統設計技能

- 資料 pipeline
 - Data ingestion from streaming sources (Kafka、MQTT、message queues)
 - ETL / ELT for time-series (清理、轉換、特徵工程)
 - Training pipeline (dataset splitting、feature pipelines)
 - Evaluation pipeline (backtest、rolling evaluation)
- 模型 pipeline
 - Model training orchestration (多實驗管理)
 - Hyperparameter tuning (Grid / Random / Bayesian)
 - Model selection / model ensemble for time series
- 部署 pipeline
 - Batch inference pipeline
 - Real-time / streaming inference pipeline
 - AB testing / shadow deployment for models

8. MLOps 與大規模部署技能

- MLOps 基本概念
 - Model versioning (模型版本管理)
 - Experiment tracking (如 MLflow, Weights & Biases)
 - Feature store (時間序列特徵管理)
 - Model registry
- 部署與監控
 - Containerization (Docker)
 - Orchestration (Kubernetes / K8s) 概念
 - REST / gRPC model serving
 - Batch vs online serving
 - Model monitoring (drift detection、performance monitoring)
 - Retraining / continual learning pipeline
- Scale-out / performance
 - Horizontal scaling (多節點服務)
 - GPU-aware serving (多 GPU 資源管理)
 - Low-latency inference (batch size / model size / quantization)

9. 即時 (Real-time) 與串流 (Streaming) 系統技能

- 流式處理框架概念
 - Kafka / Pulsar / 其他 message queue 概念
 - Stream processing (Flink, Spark Streaming, 或類似架構概念)
- 時序串流特有議題
 - Event time vs processing time
 - Late data / out-of-order data 處理
 - Windowing (tumbling / sliding / session windows)
 - State management in streaming operators
- Real-time inference
 - 低延遲預測 / 異常偵測
 - Online feature computation
 - Online learning / incremental updates (若有)

10. 機率式預測與不確定性估計技能

- 機率式 forecasting
 - Predictive distribution modeling (output full distribution not just mean)
 - Quantile regression (τ -quantile, pinball loss)
 - Prediction intervals (95% interval 等)
- 不確定性估計
 - Aleatoric vs Epistemic uncertainty
 - Bayesian neural networks (概念層級)
 - Monte Carlo dropout / ensemble methods for uncertainty
 - Calibration of probabilistic forecasts (e.g., coverage, reliability diagram)
- 評估指標
 - CRPS (Continuous Ranked Probability Score)
 - Interval coverage / width trade-off

11. Agentic AI / Agent 系統相關技能

- Agentic AI 概念
 - Agent / tool-using agent / multi-agent 系統概念
 - Orchestrating models to act as agents over data streams
- 時間序列 / 工廠情境中的 agent 能力
 - Decision-making based on forecasts / anomalies
 - Rule-based + learning-based hybrid systems
 - Event-driven workflows (agent 根據異常觸發行動)
- 實作面
 - Task planning / workflow orchestration (如 state machines, DAGs)
 - 與 backend 服務整合 (控制系統、告警系統)

12. 生產級 AI 系統與軟體工程實務

- 生產級程式設計
 - Clean code / 模組化設計
 - 單元測試 / 整合測試 (testing for models & pipelines)
 - Logging / observability
- 版本控制與團隊開發
 - Git / GitHub / GitLab workflow
 - Code review / CI/CD pipeline

- 文件與可維護性
 - API / pipeline documentation
 - Reproducible experiments (seed, config management)

13. 與製造 / 工業系統相關的 domain 技能 (技術向)

- 工廠 / 製造數據理解
 - MES / SCADA / PLC 資料流基本概念
 - OEE、產能、良率等 KPI 的時間序列行為
- 實務限制
 - Safety-critical systems 的 false positive / false negative trade-off
 - Edge vs cloud deployment (在工廠內部 vs 雲端)
 - Resource constraints (延遲、頻寬、計算資源)

如果熊熊接下來要「針對這份清單做完整複習」，建議的閱讀順序可以是：

1. 先把「時間序列理論 + 統計 + 訊號處理」補到可以直覺看懂各種序列行為 (stationary / trend / seasonality / heteroskedastic 等)。
2. 接著專攻「時間序列 DL 模型 (RNN/LSTM/TCN/Transformer)」與「forecasting / anomaly / predictive maintenance 的典型 loss 與 metrics」。
3. 然後把「pipeline + MLOps + streaming」建一兩個玩具專案 (例如 Kafka + PyTorch model 的 real-time anomaly detection)。
4. 最後再補「probabilistic forecasting / uncertainty」與「agentic AI 在製造場景的 decision-making」。

如果熊熊願意，下一步可以挑一個區塊 (例如「時間序列 DL 模型」或「probabilistic forecasting」)，請我幫你整理成「考點 + 必備定理/概念 + 可能面試問題清單」。